HANIFF HAMDAN

Software Engineer | Backend · Cloud | AI Integration

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SUMMARY

Software Engineer with 3+ years of enterprise application development experience building backend and web systems using Java/Spring Boot and C#/.NET. Experienced in REST APIs, database optimization, integration testing, cloud infrastructure, containerized environments, and CI/CD workflows. Hands-on experience with AWS, GCP, Terraform, Docker, Kubernetes/K3s, observability, IoT, and AI-enabled automation through independent engineering projects. Interested in building cloud-native web applications that integrate AI for monitoring, intelligent workflows, operational insights, and controlled automation.

EXPERIENCE

Full-Stack Developer — Heitech Padu Bhd

09/2023 – Current

Project: JPJ Teknologi Digital

- Developed and maintained Java/Spring Boot backend services supporting vehicle registration and licensing workflows for an enterprise government system.
- Worked with backend APIs and DB2 databases to support application workflows and system integrations.
- Optimized DB2 queries to improve processing efficiency and reduce data inconsistencies.
- Developed integration tests to validate API reliability and support regression testing.
- Participated in Agile sprint planning, backlog refinement, and cross-team software delivery.

Project: KPDRM System Revamp

- Redesigned and modernized legacy system components using C#/.NET and REST APIs.
- Improved SQL query performance and reduced data retrieval time by approximately 30%.
- Led UAT testing cycles, identified critical issues, and contributed to a 20% reduction in production defects.
- Developed responsive frontend components integrated with backend services.

System Developer — NIISE, IRIS Corporation Bhd

06/2023 – 09/2023

- Engineered backend APIs using C# and ASP.NET to process and transmit biometric data from fingerprint scanners in real time.
- Integrated hardware authentication systems with web applications for secure verification.
- Re-engineered immigration system modules to improve scalability and application maintainability.

Software Developer Intern — Bravonet Sdn Bhd

07/2022 – 01/2023

- Developed responsive web interfaces using JavaScript, Bootstrap, and CSS with cross-browser compatibility.
- Supported CI/CD build, test, and deployment workflows alongside senior developers.

SELECTED ENGINEERING PROJECTS

Human-in-the-Loop AI Infrastructure Monitoring Agent

Python · Prometheus · MQTT · DeepSeek API · Gemini API · Telegram

- Built a Python monitoring agent consuming Prometheus infrastructure metrics and MQTT IoT telemetry.
- Integrated LLM APIs to analyse anomalies and recommend remediation using live system context and persisted incident history.
- Implemented human-in-the-loop approval through Telegram before infrastructure actions can execute
- Built validation guardrails and an action allowlist restricting AI-generated operations to approved PM2, Docker and K3s actions.

Homelab Infrastructure & Internal Platform

K3s · Docker · Helm · Prometheus · Grafana · Loki · NestJS · React · PostgreSQL

- Operate a self-hosted K3s environment on ARM64 hardware running containerized applications and infrastructure services.
- Built observability workflows using Prometheus, Grafana, and Loki/Promtail for application and infrastructure monitoring.
- Designed and deployed an internal platform, HelmDeck, using NestJS, React/TypeScript, PostgreSQL, BullMQ, and Socket.io.
- Troubleshot K3s networking, iptables chains, container communication, and reverse-proxy CSRF configuration.
- Operate a publicly accessible Laravel environment using PHP-FPM, NGINX, Redis, MariaDB, GCP reverse proxy, and Tailscale.

Hybrid Edge-to-Cloud IoT Pipeline *— AWS, Edge Compute, Cloudflare* **| Personal Project**
Python · FastAPI · AWS IoT Core · MQTT/TLS · Terraform · S3 · CloudFront · Cloudflare

- Designed a hybrid edge-cloud IoT architecture combining Orange Pi edge compute with AWS cloud services.
- Built a Python ingestion and enrichment service that polls Tuya smart plugs, processes telemetry, calculates power-cost data, and publishes metrics to AWS IoT Core over MQTT/TLS.
- Exposed a local FastAPI backend through Cloudflare Tunnels without requiring direct port forwarding to the homelab network.
- Provisioned cloud infrastructure using Terraform for an S3-hosted web dashboard delivered through CloudFront with Origin Access Control.
- Refactored compute and database workloads from a fully serverless AWS design toward an edge-hosted model to reduce reliance on Lambda and DynamoDB.

https://d3635033e97wxx.cloudfront.net/

Laravel Web Application *– Google Cloud Platform (GCP)* **| Personal Project**
Laravel · PHP · MySQL · GCP · Docker · Redis · NGINX

- Built and deployed a Laravel web application across a local environment and a GCP VM as a hands-on hybrid cloud project.
- Implemented authentication, administrative functionality, CRUD workflows, Blade templating, authorization, validation, and transactional database operations.
- Applied Repository pattern and SOLID principles to improve application structure and maintainability.
- Improved application performance through database indexing, eager loading to address N+1 queries, and response caching using Spatie Laravel Responsecache.

http://35.230.24.195/blog/welcome-to-my-blog-aFPGQ

TECHNICAL SKILLS

- Software Engineering: **Java, Spring Boot, C#/.NET, ASP.NET, PHP/Laravel, Python, JavaScript, TypeScript, Node.js, React, Vue, NestJS, FastAPI, REST APIs**
- AI & Automation: **LLM API integration, DeepSeek API, Gemini API, AI-assisted monitoring, structured output validation, human-in-the-loop workflows, controlled automation**
- **Cloud Platforms:** AWS, Google Cloud Platform
- **AWS Services:** VPC, ECS Fargate, ECR, RDS, IAM, Lambda, DynamoDB, IoT Core, S3, CloudFront, API Gateway, SSM Parameter Store, CloudWatch
- **Infrastructure as Code:** Terraform, state management, remote backends, modules
- **Containers & Orchestration:** Docker, Kubernetes/K3s, Helm
- **CI/CD:** GitHub Actions, AWS OIDC, environment protection
- **Observability:** Prometheus, Grafana, Loki/Promtail, CloudWatch
- **Linux & Networking:** Linux administration, NGINX, MQTT, Tailscale, Cloudflare Tunnels
- **Databases:** MySQL, MariaDB, PostgreSQL, DynamoDB, DB2, SQL optimization
- **Engineering Practices:** Agile/Scrum, microservices architecture, integration testing, Test-Driven Development

EDUCATION

Bachelor of Engineering Technology, Electronic (Computer System) — Universiti Malaysia Pahang *2019 – 2023*

Diploma, Electronic Engineering (Communication System) — Politeknik Kota Bharu *2015 – 2018*

CERTIFICATIONS

- **AWS Certified Cloud Practitioner (CLF-C02)** — Amazon Web Services
- **AWS re/Start Graduate** — Amazon Web Services
- **DevOps Foundation Certification**
- JavaScript Algorithms and Data Structures